

# LeSANet: Lesion-Set-Aware In-Context Learning for Whole-Body PET-CT Tumor Segmentation

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## Abstract

Accurate segmentation of multiple metastatic lesions in whole-body PET-CT is essential for quantitative tumor burden assessment, yet remains challenging because lesions are spatially dispersed, heterogeneous in appearance, and frequently small or low-contrast. Existing in-context segmentation methods mainly encode support examples into task-level representations, which cannot reliably capture the varying cardinality and spatial configuration of metastatic lesions. We propose LeSANet, a three-dimensional lesion-aware in-context learning framework that establishes fine-grained lesion-set-level correspondences between support and query volumes. LeSANet first retrieves metabolically relevant support cases and extracts lesion prototypes to condition query features. A cross-sample interaction module then anchors these semantic prototypes to potential lesion locations in the query. During decoding, lesion-guided prompts, coarse-to-fine gated refinement, and similarity-weighted multi-support fusion preserve lesion information and progressively recover segmentation details. In addition, a size-aware loss increases the contribution of small lesions during optimization. Experiments on AutoPET-FDG, AutoPET-PSMA, and DeepPSMA demonstrate that LeSANet consistently outperforms existing medical in-context learning methods across overlap, sensitivity, precision, and boundary-distance metrics. It achieves Dice scores of 75.78%, 59.85%, and 82.58% on the three datasets, respectively, while attaining particularly strong sensitivity and HD95 performance. These results demonstrate the effectiveness of lesion-level contextual modeling for accurate and comprehensive whole-body PET-CT tumor segmentation.

## Introduction

Accurate assessment of multiple metastatic malignancies is essential for tumor staging, treatment response monitoring, and personalized therapy. This process relies on quantitative tumor burden analysis, for which accurate segmentation is a prerequisite. PET-CT combines metabolic and anatomical information and is widely used to detect and assess whole-body metastatic lesions (Zou et al. 2025). However, whole-body PET-CT segmentation remains challenging because lesions are spatially dispersed, highly heterogeneous, and often small or low contrast (Fu et al. 2025; Tasdogan et al. 2025; Sundar et al. 2022). These factors also make manual annotation difficult and demand robust automated methods.

Deep learning-based segmentation generally depends on large, well-annotated datasets and predefined tasks, making performance sensitive to training distributions and annotation quality (Rahman and Marculescu 2025; Tang et al. 2025; Liang et al. 2025; Xu et al. 2025). In clinical practice, whole-body PET-CT acquisition is costly and annotation requires substantial expert effort, resulting in scarce and unevenly distributed labels (Tarai et al. 2025). Moreover, patient variability, scanning differences, and diverse lesion appearances further limit generalization (Xia et al. 2025). In-context learning (ICL) offers a flexible alternative: given a few annotated support image-mask pairs, it segments a query image without extensive retraining. General vision models such as SegGPT (Wang et al. 2023b) and medical ICL methods such as Iris (Gao et al. 2025) have demonstrated its potential for low-annotation and distribution-shift settings.

Nevertheless, existing ICL methods are not tailored to multiple metastatic lesions in whole-body PET-CT. They typically rely on task-level correspondence, assuming that support examples provide stable task semantics (Bougourzi and Hadid 2025). This assumption is unreliable for sparse, spatially dispersed lesions with substantial variation in size, location, and metabolic activity, as a single support example rarely captures all lesion patterns (Li et al. 2021). Consequently, global feature aggregation may dilute fine-grained lesion information, favoring salient lesions while missing small or weak-signal lesions (Gatidis et al. 2024). Moreover, conventional overlap-based objectives are insufficiently sensitive to missed small lesions (Kofler et al. 2022).

To address these limitations, we propose LeSANet, a lesion-aware ICL framework for whole-body PET-CT tumor segmentation. LeSANet shifts correspondence modeling from the task level to the lesion-set level, explicitly establishing fine-grained relationships between potential lesions in support and query images. This lesion-level context improves the identification of heterogeneous, dispersed, and small lesions, enabling more comprehensive whole-body metastatic lesion segmentation.

The main contributions of this work are summarized as follows:

1. We propose LeSANet, a 3D lesion-aware in-context learning framework for segmenting heterogeneous, spatially dispersed, and multi-scale metastatic lesions in whole-body

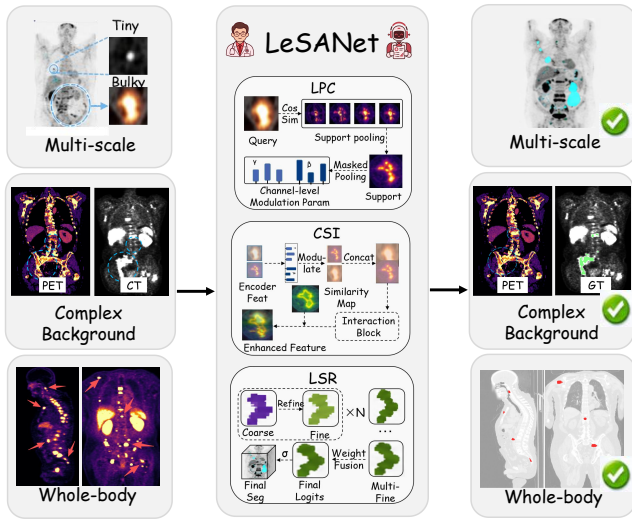


Figure 1: Overview of LeSANet, which leverages lesion prior correspondence and refinement to achieve robust whole-body PET-CT segmentation under multi-scale lesions and complex backgrounds.

PET-CT.

2. We introduce a retrieval-enhanced lesion semantic prototype and spatial correspondence mechanism, transferring lesion priors from semantically similar support cases to improve the representation and localization of low-contrast and dispersed lesions.

3. We develop cross-case lesion interaction and prompt-guided decoding to fuse complementary support-query information and progressively refine lesion boundaries.

4. We propose a size-aware loss that increases the optimization emphasis on small lesions, improving sensitivity to tiny and sparse metastases.

5. Experiments on AutoPET-FDG, AutoPET-PSMA, and DeepPSMA demonstrate competitive performance in Dice, PPV, Sensitivity, and HD95, particularly for small and multiple lesions.

## Related Work

### Whole-Body PET-CT Tumor Segmentation

The autoPET challenge series (Gatidis et al. 2024) established a public benchmark for whole-body PET-CT tumor segmentation. Mainstream approaches are primarily based on nnU-Net (Isensee et al. 2021) and its variants, often combined with false-positive suppression post-processing to address erroneous detections caused by physiologically high-uptake tissues, such as the brain and bladder. Challenge result analyses have consistently identified missed small and low-uptake lesions as a major failure mode. Common strategies that remove small connected components further sacrifice these lesions while reducing false positives. This issue fundamentally arises from the optimization objective: region-overlap losses such as Dice are voxel-weighted, and metastatic lesions containing only tens of voxels contribute

negligibly to the gradients. Moreover, these models are task-specific and require retraining when deployed across institutions or tracer types.

### Visual In-Context Segmentation

ICL originated in natural language processing (Brown et al. 2020). Painter (Wang et al. 2023a) first introduced this paradigm to computer vision by formulating dense prediction as mask completion on a concatenated canvas, where task semantics are implicitly encoded in the input-output mappings of support examples. SegGPT (Wang et al. 2023b) further unified segmentation as in-context coloring. Its random color mappings force the model to rely on contextual information rather than memorizing fixed associations between semantic categories and colors. INSID3 (Cuttano et al. 2026) showed that frozen DINOv3 dense features can support training-free in-context segmentation through semantic matching between support and query images.

These methods were developed for natural images, where targets are typically single, salient, and spatially connected. Their correspondences are therefore established at the object or part category level. In contrast, the target in whole-body PET-CT is a sparse lesion set whose cardinality and spatial configuration vary across patients. Category-level correspondence is insufficient to characterize this setting.

### In-Context Segmentation for Medical Imaging

UniverSeg (Butoi et al. 2023) employs CrossBlocks to individually interact the query with each support example and then averages the resulting representations, achieving permutation-invariant context fusion. Tyche (Rakic et al. 2024) introduces stochastic inference to generate multiple candidate segmentations and model annotation ambiguity. Iris (Gao et al. 2025) distills reference image-label pairs into compact task embeddings and decouples task encoding from query inference. EICSeg (Xie et al. 2026) reformulates contextual interaction as explicit retrieval and leverages vision foundation models to capture dense correspondences between queries and prompts. Neuroverse (Hu et al. 2025) and Medverse (Hu et al. 2026) extend ICL to three-dimensional imaging. They alleviate memory and resolution bottlenecks through adaptive parallel-serial processing and next-scale autoregression, respectively.

The context representations in these methods are all obtained through compression and are thus invariant to target cardinality and spatial configuration. This prior is appropriate for unique organs with stable locations, but it discards information essential for characterizing multiple metastatic lesions in whole-body PET-CT.

## Method

### Framework Overview

Whole-body PET-CT lesions exhibit substantial variation in scale and are spatially dispersed. Consequently, randomly sampled support images may differ markedly from the query in metabolic patterns and fail to provide reliable lesion priors. To address this issue, we propose LeSANet, a three-

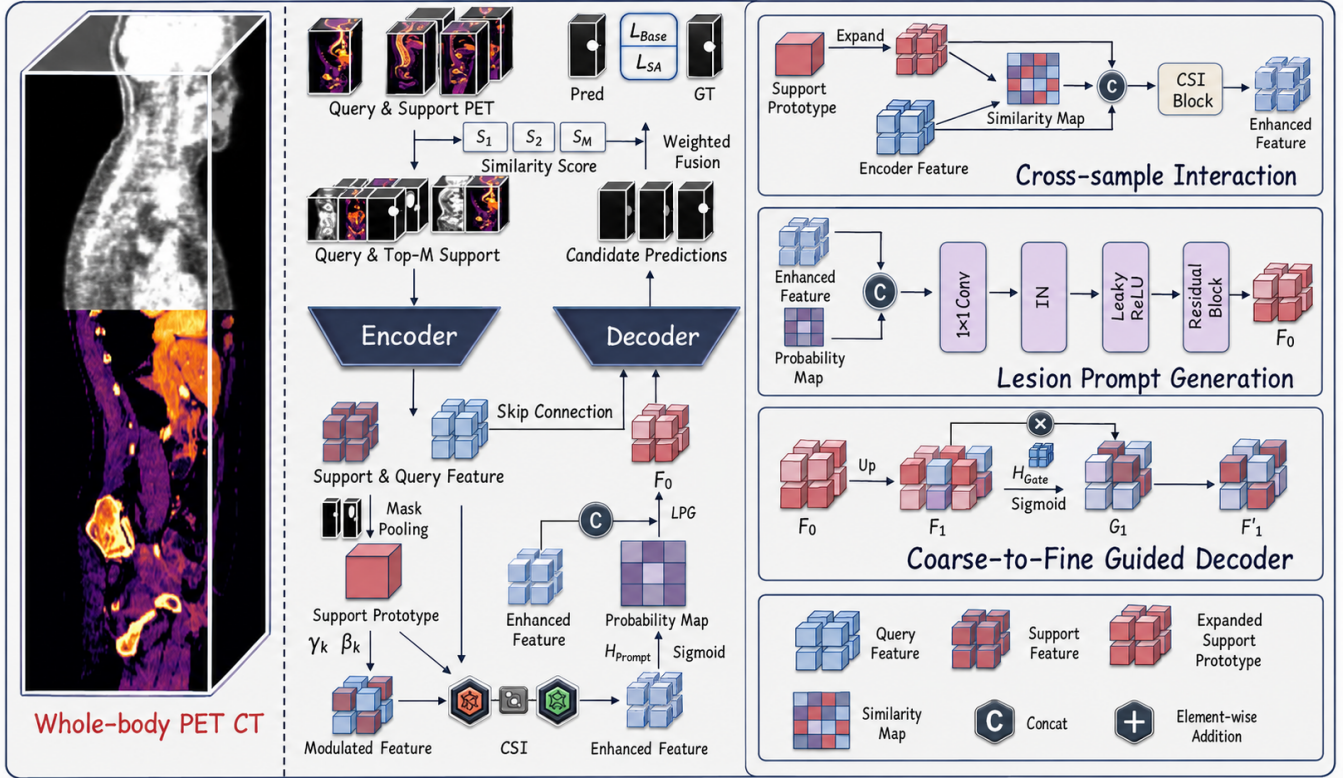


Figure 2: Overview of the proposed LeSANet framework for whole-body PET-CT lesion segmentation. Given a query image, semantically similar support cases are retrieved to construct lesion prototypes. Cross-sample interaction transfers lesion priors from support to query features, while lesion prompt generation and a coarse-to-fine guided decoder progressively enhance lesion representations and refine segmentation predictions. At inference time, the retrieval candidates are restricted to a fixed pool of eight labeled training volumes. For each query, SR ranks these candidates and selects the top- $M$  supports, where  $M = 2$ . Thus, no test-set image or annotation is used as a support example.

dimensional in-context learning segmentation framework (Fig. 2).

Given a query image and a support pool, the lesion prior construction (LPC) module first retrieves the top- $M$  support samples with metabolic patterns most similar to that of the query. It then extracts semantic prototypes from their lesion regions to condition the query features (Section 3.2). Subsequently, the query and support images are fed into a shared three-dimensional residual encoder to extract multi-scale features. At the bottleneck layer, the cross-sample interaction (CSI) module establishes lesion-set-level correspondences, anchoring the lesion priors to the spatial locations in the query image (Section 3.3). The lesion-guided segmentation refinement (LSR) module propagates lesion prompts throughout the decoding path, enabling progressive recovery from coarse localization to fine boundary delineation (Section 3.4).

These operations are performed in parallel for the  $M$  support samples, producing  $M$  candidate segmentations. The candidates are adaptively fused into the final prediction according to query-support similarity. During training, we further introduce a size-aware loss that dynamically allocates

supervision strength according to lesion volume, preventing gradients from being dominated by a small number of large lesions (Section 3.5).

### Lesion Prior Construction

The effectiveness of in-context learning depends critically on the quality of priors provided by the support samples. However, metabolic patterns and lesion distributions vary substantially across individuals in whole-body PET-CT, making randomly sampled supports often semantically irrelevant to the query. This module retrieves support samples that are most similar to the query through support retrieval and extracts discriminative lesion prototypes via support prototype conditioning, thereby providing reliable priors for subsequent lesion-set-level correspondence modeling.

**Support Retrieval** Because the PET channel directly reflects lesion metabolic activity, we construct a case-level metabolic descriptor using PET images only. Given  $X_{\text{PET}}$ , we sequentially apply non-negative clipping and percentile thresholding to emphasize high-uptake hotspots, downsample the image to a fixed size  $S$  to eliminate differences in

image volume, and then flatten and  $\ell_2$ -normalize it:

$$\mathbf{d} = \text{Norm}(\text{vec}(\text{Down}_S(\text{Thresh}_p(\text{ReLU}(\mathbf{X}_{\text{PET}}))))). \quad (1)$$

The metabolic similarity between the query and the  $k$ -th support sample is computed using cosine similarity. The top- $M$  samples are then selected from the support pool to form the support set  $\mathcal{S}$ :

$$s_k = \text{Sim}(\mathbf{d}^q, \mathbf{d}_k^s), \quad \mathcal{S} = \text{Top-}M\left(\{s_k\}_{k=1}^K\right). \quad (2)$$

Here,  $\text{ReLU}(\cdot)$  denotes non-negative clipping;  $\text{Thresh}_p(\cdot)$  retains high-uptake regions above the  $p$ -th percentile;  $\text{Down}_S(\cdot)$  denotes downsampling;  $\text{vec}(\cdot)$  denotes vectorization; and  $\text{Norm}(\cdot)$  denotes  $\ell_2$  normalization. Retrieval is performed only once before the forward pass and therefore incurs negligible computational overhead. The similarity score  $s_k$  is reused as a fusion weight in Section 3.4.3.

**Support Prototype Conditioning** Although retrieval improves the overall matching between the Query and Support samples, Support features still contain substantial background responses unrelated to lesions. To extract discriminative lesion semantics, SPC performs masked average pooling on the bottleneck features of the  $k$ -th Support sample using its lesion mask, producing a lesion semantic prototype:

$$\mathbf{p}_k = \frac{\sum_i \mathbf{M}_{k,i}^s \odot \mathbf{F}_{k,i}^s}{\sum_i \mathbf{M}_{k,i}^s + \epsilon}, \quad (3)$$

where  $i$  indexes spatial locations. The prototype is subsequently passed through a projection network  $\phi(\cdot)$ , consisting of two fully connected layers and Layer Normalization, to reduce redundancy and obtain a compact representation:

$$\hat{\mathbf{p}}_k = \phi(\mathbf{p}_k), \quad (4)$$

A parameter prediction network  $\psi(\cdot)$  then generates channel-wise modulation parameters to condition and enhance the Query feature  $\mathbf{F}^q$ :

$$(\gamma_k, \beta_k) = \psi(\hat{\mathbf{p}}_k), \quad \tilde{\mathbf{F}}_k^q = \gamma_k \odot \mathbf{F}^q + \beta_k. \quad (5)$$

Here,  $\odot$  denotes element-wise multiplication, and  $\gamma_k$  and  $\beta_k$  are broadcast along the spatial dimensions. Compared with directly concatenating Support features, conditional modulation recalibrates the channel responses of the Query feature according to the lesion prototype. This design suppresses background interference and amplifies lesion-relevant channels with negligible additional parameter overhead.

### Cross-Sample Interaction

The prototype  $\hat{\mathbf{p}}_k$  is global and can only characterize what semantic pattern a lesion exhibits; it cannot determine where the lesion is located in the Query. CSI addresses this limitation by establishing lesion-set-level spatial correspondences. Specifically, the prototype is first broadcast into a constant feature map with the same spatial dimensions as the Query feature. Channel-wise normalized cosine similarity is then

computed at each spatial location to obtain a lesion similarity map:

$$\begin{aligned} \mathbf{P}_k^b &= \text{Broadcast}(\hat{\mathbf{p}}_k), \\ \mathbf{A}_{k,i} &= \frac{\tilde{\mathbf{F}}_{k,i}^q \cdot \mathbf{P}_{k,i}^b}{\|\tilde{\mathbf{F}}_{k,i}^q\|_2 \|\mathbf{P}_{k,i}^b\|_2 + \epsilon}. \end{aligned} \quad (6)$$

Here,  $\mathbf{A}_{k,i}$  quantifies the correspondence between the  $i$ -th spatial location in the Query and the lesion prior provided by the  $k$ -th Support sample. A higher response indicates that the corresponding location is more likely to belong to the target lesion. The Query feature, broadcast prototype, and similarity map are concatenated along the channel dimension and fed into the cross-sample interaction unit  $\mathcal{F}_{\text{CSI}}(\cdot)$ , which consists of a  $1 \times 1 \times 1$  convolution, Instance Normalization, LeakyReLU, and residual blocks. The enhanced feature is obtained through a residual connection:

$$\mathbf{F}_k^{\text{CSI}} = \tilde{\mathbf{F}}_k^q + \mathcal{F}_{\text{CSI}}\left(\left[\tilde{\mathbf{F}}_k^q; \mathbf{P}_k^b; \mathbf{A}_k\right]\right). \quad (7)$$

### Lesion-guided Segmentation Refinement

After LPC and CSI, lesion priors have been injected into the bottleneck feature of the Query. However, a conventional Decoder progressively dilutes high-level semantic information during successive upsampling stages, causing small lesions and lesions with ambiguous boundaries to be easily lost. LSR preserves lesion priors throughout the decoding pathway through lesion prompt generation, coarse-to-fine gated decoding, and adaptive multi-Support fusion, thereby progressively recovering lesion details for more accurate segmentation.

**Lesion Prompt Generation** A coarse lesion probability map is generated from the enhanced bottleneck feature to provide high-level semantic guidance for the decoding process:

$$\mathbf{P}_k^{\text{coarse}} = \sigma(\mathcal{F}_{\text{Fuse}}(\mathbf{F}_k^{\text{CSI}})), \quad (8)$$

where  $\sigma(\cdot)$  denotes the Sigmoid activation function, and  $\mathcal{F}_{\text{Fuse}}(\cdot)$  consists of a  $1 \times 1 \times 1$  convolution, Instance Normalization, LeakyReLU, and residual blocks. This design provides the Decoder with coarse lesion localization information before spatial resolution recovery, enabling it to continuously focus on potential lesion regions during subsequent decoding.

**Coarse-to-Fine Guided Decoding** The Decoder employs four upsampling stages, with gating strategies introduced at the two intermediate stages. Let  $\mathbf{F}_l = \text{Up}^l(\mathbf{F}_{l-1})$  denote the decoded feature after the  $l$ -th upsampling operation, and let  $\mathbf{E}_l$  denote the Encoder skip-connection feature at the corresponding scale. The gating map is generated by a gated segmentation head and jointly modulates both the decoded feature and the skip-connection feature:

$$\begin{aligned} \mathbf{G}_l &= \sigma(\mathcal{H}_l^{\text{gate}}(\mathbf{F}_l)), \\ \tilde{\mathbf{F}}_l &= \mathbf{G}_l \odot \mathbf{F}_l, \\ \tilde{\mathbf{E}}_l &= \mathbf{G}_l \odot \mathbf{E}_l. \end{aligned} \quad (9)$$

Here,  $l = 1$  and  $l = 2$  correspond to coarse localization guidance and fine boundary guidance, respectively. Unlike conventional Decoders that pass spatial information passively through skip connections, the gating mechanism enables lesion priors to actively participate in feature fusion at each decoding stage. It amplifies lesion responses while suppressing background interference, achieving progressive refinement from coarse localization to detailed boundary recovery.

**Multi-Support Fusion** The matching quality between different Support samples and the Query varies, leading to different qualities of their corresponding candidate segmentations. Simple averaging may cause low-quality Support samples to dilute the contribution of high-quality priors. Therefore, the similarity score  $s_k$  from the SR stage is reused to compute fusion weights via temperature-scaled Softmax, and the  $M$  candidate predictions are aggregated by weighted summation:

$$w_k = \frac{\exp(s_k/\tau)}{\sum_{j=1}^M \exp(s_j/\tau)}, \quad \mathbf{P}^{\text{final}} = \sum_{k=1}^M w_k \mathbf{P}_k^{\text{pred}}. \quad (10)$$

The temperature coefficient  $\tau$  controls the smoothness of the weight distribution. A smaller  $\tau$  makes the model rely more strongly on the Support sample most similar to the Query, whereas a larger  $\tau$  produces more balanced contributions across Support samples. Without introducing additional parameters, this strategy emphasizes high-quality priors and suppresses interference from low-quality Support samples, improving prediction stability and robustness.

### Size-aware Loss

In whole-body PET-CT, lesion volumes vary substantially, causing standard Dice and BCE losses to be dominated by large lesions and leaving small lesions underfitted. We therefore propose a Size-aware Loss that dynamically weights supervision according to lesion volume (Algorithm 1).

At each iteration, connected components are extracted from the ground-truth mask, with components smaller than  $v_{\min}$  removed as noise. Each remaining lesion is assigned an inverse-volume weight normalized by  $s_{\text{ref}}$ , scaled by  $w_{\text{scale}}$ , and clipped at  $w_{\text{max}}$ . These lesion weights form a voxel-wise map  $W_{\text{sa}}$  for computing the weighted BCE loss  $L_{\text{sa}}$ .

The final objective combines  $L_{\text{sa}}$  with standard Dice and cross-entropy losses using  $\lambda_{\text{sa}}$ ,  $\lambda_{\text{dice}}$ , and  $\lambda_{\text{ce}}$ . By shifting supervision toward medium and small lesions, this loss improves their detection while preserving stable segmentation of large lesions.

## Experiment

### Datasets

AutoPET-FDG (Gatidis et al. 2022) comprises 1,014  $^{18}\text{F}$ -FDG PET-CT scans acquired at the University Hospital of Tübingen using a Siemens Biograph mCT scanner under consistent protocols. It includes 501 patients with malignant

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### Algorithm 1: Size-Aware Loss

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**Require:** Training set  $\mathcal{D}$ , model  $f(\cdot; W)$   
**Require:**  $\lambda_{\text{dice}}, \lambda_{\text{ce}}, \lambda_{\text{sa}}, s_{\text{ref}}, w_{\text{scale}}, w_{\text{max}}, v_{\min}$

- 1: **for**  $epoch = 1$  to  $E$  **do**
- 2:   **for all**  $(x, y) \in \mathcal{D}$  **do**
- 3:      $\hat{y} \leftarrow f(x; W)$  {Forward propagation}
- 4:      $L_{\text{dice}} \leftarrow \text{DiceLoss}(\hat{y}, y)$
- 5:      $L_{\text{ce}} \leftarrow \text{CrossEntropy}(\hat{y}, y)$
- 6:     Extract connected components  $\{C_k\}$  from  $y$
- 7:     **for all**  $C_k$  **do**
- 8:       Compute lesion volume  $|C_k|$
- 9:       **if**  $|C_k| < v_{\min}$  **then**
- 10:         **continue**
- 11:       **end if**
- 12:        $s_k \leftarrow |C_k|/s_{\text{ref}}$
- 13:        $w_k \leftarrow \min(w_{\text{max}}, w_{\text{scale}}/s_k)$
- 14:     **end for**
- 15:     Build voxel-wise weight map  $W_{\text{sa}}$
- 16:     Compute  $L_{\text{sa}}$  using  $W_{\text{sa}}, \hat{y}$ , and  $y$
- 17:      $L \leftarrow \lambda_{\text{dice}}L_{\text{dice}} + \lambda_{\text{ce}}L_{\text{ce}} + \lambda_{\text{sa}}L_{\text{sa}}$
- 18:     Update model parameters  $W$
- 19:   **end for**
- 20: **end for**

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melanoma, lymphoma, or lung cancer and 513 negative controls, constituting an in-domain benchmark.

AutoPET-PSMA (Jeblick et al. 2026) comprises pre-treatment and post-treatment PSMA PET-CT scans from 597 patients with prostate cancer at the University Hospital of Munich, including 537 patients with PSMA-avid lesions and 60 without lesions. Data were acquired using three Siemens and GE scanners, introducing shifts in tracer, center, and scanner relative to AutoPET-FDG.

DeepPSMA (Jackson et al. 2025) includes paired PSMA and FDG PET-CT scans from 100 patients with metastatic castration-resistant prostate cancer before  $^{177}\text{Lu}$ -PSMA therapy. Tracer-specific whole-body tumor masks were derived using an  $\text{SUV} \geq 3$  threshold for PSMA and an adaptive liver-based threshold for FDG, followed by manual removal of physiological uptake. The dataset further differs in annotation protocol and predominantly contains bone metastases.

### Data Processing

All datasets were obtained from their official releases in NIfTI format. PET images were converted to standardized uptake values (SUVs), and PET and CT volumes were resampled to an isotropic resolution. CT intensities were clipped to  $[-100, 200]$  HU, after which PET and CT volumes were independently normalized using volume-wise Z-score normalization. For each model, the patch size followed the setting specified in the original publication or its recommended configuration and was kept unchanged during training, validation, and testing.

At the patient level, each dataset was divided into two disjoint cohorts,  $A$  and  $B$ , of approximately equal size. Cohort  $A$  was further split into  $A_{\text{train}}$  and  $A_{\text{val}}$  at a ratio of 8:2,



Table 1: **Quantitative comparison of 3D medical image segmentation performance on the FDG, PSMA, and Deep PSMA datasets.** Dice, PPV, and Sensitivity are reported in percentage (%); HD95 is a distance-based metric.  $\uparrow$  indicates higher is better, and  $\downarrow$  indicates lower is better. Fully supervised methods are included as upper-bound references. All ICL methods use two support volumes per query, selected from a shared pool of eight labeled training cases; all methods are trained for 200 epochs on the corresponding dataset.

| Method                                   | FDG                                 |                                     |                                     |                                     | PSMA                                |                                     |                                     |                                     | Deep PSMA                           |                                     |                                     |                                    |
|------------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|------------------------------------|
|                                          | Dice $\uparrow$                     | PPV $\uparrow$                      | Sens. $\uparrow$                    | HD95 $\downarrow$                   | Dice $\uparrow$                     | PPV $\uparrow$                      | Sens. $\uparrow$                    | HD95 $\downarrow$                   | Dice $\uparrow$                     | PPV $\uparrow$                      | Sens. $\uparrow$                    | HD95 $\downarrow$                  |
| <i>Fully Supervised Upper Bound</i>      |                                     |                                     |                                     |                                     |                                     |                                     |                                     |                                     |                                     |                                     |                                     |                                    |
| nnU-Net                                  | <u>69.61 <math>\pm</math> 28.63</u> | 69.82 $\pm$ 30.67                   | 75.40 $\pm$ 30.18                   | <u>38.85 <math>\pm</math> 68.24</u> | 54.40 $\pm$ 30.41                   | 61.81 $\pm$ 33.12                   | 55.74 $\pm$ 33.38                   | 44.57 $\pm$ 69.01                   | <u>88.55 <math>\pm</math> 4.26</u>  | <u>88.55 <math>\pm</math> 9.99</u>  | <u>89.84 <math>\pm</math> 6.01</u>  | <u>8.43 <math>\pm</math> 11.49</u> |
| MedFormer                                | 68.58 $\pm$ 19.99                   | <u>71.30 <math>\pm</math> 20.96</u> | <u>75.83 <math>\pm</math> 23.38</u> | 54.10 $\pm$ 71.26                   | <u>65.53 <math>\pm</math> 20.38</u> | <u>67.51 <math>\pm</math> 20.62</u> | <u>71.88 <math>\pm</math> 25.78</u> | <u>32.84 <math>\pm</math> 54.21</u> | 81.68 $\pm$ 4.18                    | 81.36 $\pm$ 6.00                    | 82.46 $\pm$ 5.92                    | <u>2.60 <math>\pm</math> 2.13</u>  |
| SwinUNETR                                | <u>81.31 <math>\pm</math> 13.74</u> | <u>85.37 <math>\pm</math> 13.25</u> | <u>82.47 <math>\pm</math> 20.14</u> | <u>21.05 <math>\pm</math> 34.64</u> | <u>70.08 <math>\pm</math> 18.79</u> | <u>72.55 <math>\pm</math> 20.75</u> | <u>72.75 <math>\pm</math> 22.79</u> | <u>23.38 <math>\pm</math> 32.50</u> | <u>89.75 <math>\pm</math> 4.41</u>  | <u>91.66 <math>\pm</math> 8.51</u>  | <u>88.92 <math>\pm</math> 7.03</u>  | 16.97 $\pm$ 32.59                  |
| <i>In-Context Learning (ICL) Methods</i> |                                     |                                     |                                     |                                     |                                     |                                     |                                     |                                     |                                     |                                     |                                     |                                    |
| Iris                                     | <u>67.15 <math>\pm</math> 26.06</u> | <u>75.11 <math>\pm</math> 28.36</u> | 68.11 $\pm$ 29.65                   | 8.92 $\pm$ 8.83                     | 49.33 $\pm$ 32.19                   | <u>56.07 <math>\pm</math> 36.27</u> | 55.69 $\pm$ 34.85                   | 11.70 $\pm$ 11.82                   | <u>76.91 <math>\pm</math> 16.85</u> | <u>81.25 <math>\pm</math> 23.54</u> | 78.17 $\pm$ 6.06                    | 10.04 $\pm$ 10.74                  |
| Medverse                                 | 57.90 $\pm$ 26.77                   | 51.32 $\pm$ 27.97                   | <u>82.47 <math>\pm</math> 24.43</u> | 15.57 $\pm$ 13.79                   | <u>51.26 <math>\pm</math> 23.62</u> | 53.28 $\pm$ 28.37                   | <u>64.77 <math>\pm</math> 28.10</u> | 14.49 $\pm$ 12.26                   | 53.79 $\pm$ 18.44                   | 43.36 $\pm$ 19.67                   | <u>84.29 <math>\pm</math> 14.03</u> | 16.69 $\pm$ 9.85                   |
| Neuroverse                               | 48.70 $\pm$ 27.95                   | 58.98 $\pm$ 31.90                   | 49.44 $\pm$ 29.83                   | <u>7.37 <math>\pm</math> 5.82</u>   | 42.53 $\pm$ 22.87                   | 49.95 $\pm$ 30.02                   | 45.57 $\pm$ 26.54                   | <u>7.22 <math>\pm</math> 5.21</u>   | 64.33 $\pm$ 21.98                   | 71.01 $\pm$ 23.51                   | 65.55 $\pm$ 21.60                   | 6.43 $\pm$ 5.23                    |
| Tyche                                    | 39.01 $\pm$ 34.04                   | 43.37 $\pm$ 38.09                   | 41.94 $\pm$ 39.22                   | 15.10 $\pm$ 17.98                   | 45.62 $\pm$ 36.49                   | 50.49 $\pm$ 41.98                   | 54.42 $\pm$ 42.03                   | 7.25 $\pm$ 16.92                    | 67.29 $\pm$ 31.51                   | 64.07 $\pm$ 33.51                   | 77.02 $\pm$ 28.63                   | <u>4.84 <math>\pm</math> 10.83</u> |
| UniverSeg                                | 18.03 $\pm$ 20.15                   | 15.14 $\pm$ 21.86                   | 46.89 $\pm$ 43.56                   | 33.25 $\pm$ 16.69                   | 20.21 $\pm$ 34.60                   | 24.78 $\pm$ 41.34                   | 20.04 $\pm$ 35.73                   | 8.41 $\pm$ 12.37                    | 47.48 $\pm$ 37.37                   | 53.31 $\pm$ 41.73                   | 49.70 $\pm$ 35.39                   | 9.69 $\pm$ 15.42                   |
| LeSASNet                                 | <u>75.78 <math>\pm</math> 20.24</u> | <u>78.88 <math>\pm</math> 22.35</u> | <u>82.53 <math>\pm</math> 21.19</u> | <u>5.08 <math>\pm</math> 6.52</u>   | <u>59.85 <math>\pm</math> 30.56</u> | <u>62.94 <math>\pm</math> 33.77</u> | <u>65.76 <math>\pm</math> 33.78</u> | <u>6.92 <math>\pm</math> 15.14</u>  | <u>82.58 <math>\pm</math> 9.43</u>  | <u>81.86 <math>\pm</math> 20.52</u> | <u>89.65 <math>\pm</math> 7.30</u>  | <u>4.21 <math>\pm</math> 6.52</u>  |

Table 2: Quantitative ablation study on the **FDG, PSMA, and Deep PSMA datasets**. The component matrix reports the inclusion of SR, SPC, CSI, LSR, and  $L_{SA}$ . Higher values are better for Dice, PPV, and Sensitivity, while lower values are better for HD95. The best results are shown in **bold**, and the runner-ups are underlined. We evaluate  $M \in \{1, 2, 4\}$  while keeping the support pool fixed. We select  $M = 2$  as the default because it provides a favorable trade-off between segmentation accuracy and inference cost.

| Components   |              |              |              |              |  | FDG                                 |                                     |                                     |                                   | PSMA                                |                                     |                                     |                                    | Deep PSMA                          |                                     |                                    |                                   |
|--------------|--------------|--------------|--------------|--------------|--|-------------------------------------|-------------------------------------|-------------------------------------|-----------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|------------------------------------|------------------------------------|-------------------------------------|------------------------------------|-----------------------------------|
| SR           | SPC          | CSI          | LSR          | $L_{SA}$     |  | Dice $\uparrow$                     | PPV $\uparrow$                      | Sens. $\uparrow$                    | HD95 $\downarrow$                 | Dice $\uparrow$                     | PPV $\uparrow$                      | Sens. $\uparrow$                    | HD95 $\downarrow$                  | Dice $\uparrow$                    | PPV $\uparrow$                      | Sens. $\uparrow$                   | HD95 $\downarrow$                 |
| $\times$     | $\times$     | $\times$     | $\times$     | $\times$     |  | 73.05 $\pm$ 21.64                   | 74.38 $\pm$ 25.02                   | 78.79 $\pm$ 23.94                   | 8.37 $\pm$ 9.24                   | 57.93 $\pm$ 32.33                   | 60.31 $\pm$ 34.21                   | 62.05 $\pm$ 34.20                   | 9.93 $\pm$ 13.24                   | 77.70 $\pm$ 13.32                  | 74.06 $\pm$ 20.87                   | 87.60 $\pm$ 7.82                   | 7.08 $\pm$ 8.58                   |
| $\checkmark$ | $\times$     | $\times$     | $\times$     | $\times$     |  | 73.20 $\pm$ 20.36                   | <u>77.85 <math>\pm</math> 22.18</u> | 79.51 $\pm$ 22.83                   | 8.19 $\pm$ 9.11                   | 59.13 $\pm$ 31.01                   | 60.95 $\pm$ 33.44                   | 61.69 $\pm$ 33.14                   | 8.83 $\pm$ 11.97                   | 78.21 $\pm$ 14.15                  | 75.64 $\pm$ 23.76                   | <u>88.87 <math>\pm</math> 8.17</u> | 7.40 $\pm$ 8.84                   |
| $\checkmark$ | $\checkmark$ | $\times$     | $\times$     | $\times$     |  | 74.13 $\pm$ 23.52                   | 73.35 $\pm$ 26.44                   | <u>80.83 <math>\pm</math> 23.95</u> | 6.45 $\pm$ 7.86                   | 59.53 $\pm$ 31.09                   | <u>64.53 <math>\pm</math> 33.93</u> | 59.77 $\pm$ 32.94                   | 8.70 $\pm$ 11.72                   | 80.21 $\pm$ 11.77                  | 81.28 $\pm$ 17.96                   | 83.81 $\pm$ 10.31                  | 7.23 $\pm$ 8.73                   |
| $\checkmark$ | $\checkmark$ | $\checkmark$ | $\times$     | $\times$     |  | 74.63 $\pm$ 21.10                   | 73.08 $\pm$ 24.17                   | 74.66 $\pm$ 24.71                   | 7.34 $\pm$ 7.44                   | 59.70 $\pm$ 32.17                   | 60.05 $\pm$ 34.14                   | 63.75 $\pm$ 34.01                   | 9.14 $\pm$ 11.43                   | 81.18 $\pm$ 10.59                  | 80.94 $\pm$ 19.56                   | 86.01 $\pm$ 7.67                   | 7.11 $\pm$ 8.76                   |
| $\checkmark$ | $\checkmark$ | $\checkmark$ | $\checkmark$ | $\times$     |  | <u>75.01 <math>\pm</math> 21.41</u> | 76.31 $\pm$ 24.28                   | 79.26 $\pm$ 22.23                   | 6.39 $\pm$ 6.82                   | <u>59.74 <math>\pm</math> 31.12</u> | 61.72 $\pm$ 33.25                   | <u>64.15 <math>\pm</math> 33.12</u> | <u>7.53 <math>\pm</math> 11.51</u> | <u>82.34 <math>\pm</math> 8.67</u> | <u>81.64 <math>\pm</math> 17.24</u> | 86.90 $\pm$ 8.17                   | <u>6.65 <math>\pm</math> 8.66</u> |
| $\times$     | $\times$     | $\times$     | $\times$     | $\checkmark$ |  | 73.21 $\pm$ 21.19                   | 73.38 $\pm$ 25.17                   | 80.29 $\pm$ 21.75                   | <u>6.31 <math>\pm</math> 7.99</u> | 58.12 $\pm$ 31.37                   | 62.80 $\pm$ 33.11                   | 60.60 $\pm$ 33.85                   | 9.35 $\pm$ 11.86                   | 78.53 $\pm$ 15.69                  | 75.02 $\pm$ 23.25                   | 87.94 $\pm$ 6.67                   | 6.91 $\pm$ 8.89                   |
| $\checkmark$ | $\checkmark$ | $\checkmark$ | $\checkmark$ | $\checkmark$ |  | <u>75.78 <math>\pm</math> 20.24</u> | <u>78.88 <math>\pm</math> 22.35</u> | <u>82.53 <math>\pm</math> 21.19</u> | <u>5.08 <math>\pm</math> 6.52</u> | <u>59.85 <math>\pm</math> 30.56</u> | <u>62.94 <math>\pm</math> 33.77</u> | <u>65.76 <math>\pm</math> 33.78</u> | <u>6.92 <math>\pm</math> 15.14</u> | <u>82.58 <math>\pm</math> 9.43</u> | <u>81.86 <math>\pm</math> 20.52</u> | <u>89.65 <math>\pm</math> 7.30</u> | <u>4.21 <math>\pm</math> 6.52</u> |

whereas cohort  $B$  was split into  $B_{\text{train}}$ ,  $B_{\text{val}}$ , and  $B_{\text{test}}$  at a ratio of 4:2:4. All scans from the same patient were assigned to the same subset to prevent patient-level data leakage.

The context-aware models were trained on  $A_{\text{train}}$ , validated on  $A_{\text{val}}$ , and evaluated on  $B_{\text{test}}$ , using  $N$  annotated support images sampled from  $A_{\text{train}}$ . The fully supervised models were trained on  $A_{\text{train}} \cup B_{\text{train}}$ , validated on  $B_{\text{val}}$ , and evaluated on the same held-out  $B_{\text{test}}$ . All models were trained for 200 epochs and evaluated once using the checkpoint with the highest validation Dice score. For AutoPET-FDG, the 513 negative-control cases contained no annotated tumor lesions and were therefore excluded from the calculation of Dice, PPV, Sensitivity, and HD95. Consequently, these metrics were computed only over lesion-positive cases.

### Implementation Details

All methods were trained on the training split of the target dataset for 200 epochs using the same optimization schedule. During inference, each query volume was paired with  $M = 2$  support volumes. The support pool consisted of eight labeled volumes randomly sampled from the training set and remained fixed for all compared methods. The support annotations were therefore always obtained from the training split and were available at inference time, following the standard in-context segmentation setting. No images or annotations from the test split were used as support samples.

For a fair comparison, all methods used the same number of support volumes ( $M = 2$ ), the same support pool size (eight training volumes), and the same training/testing data splits. Each method selected its support samples using its original or method-specific selection strategy. In particular, LeSA-Net uses the proposed Similarity Retrieval (SR) module to retrieve the two most relevant supports from the common pool, whereas competing methods follow their respective support-selection mechanisms.

All models were optimized using AdamW with an initial learning rate of  $1 \times 10^{-4}$  and a weight decay of  $1 \times 10^{-5}$ . A cosine-annealing learning-rate scheduler with  $T_{\text{max}} = 50$  was applied after each epoch, with the minimum learning rate set to  $1 \times 10^{-6}$ . For every model, the checkpoint achieving the highest Dice score on the corresponding validation set was retained for final evaluation.

### Competitive Experiments

Table 1 present the quantitative results of LeSANet on AutoPET-FDG, AutoPET-PSMA, and DeepPSMA. LeSANet outperforms existing ICL methods in Dice, PPV, sensitivity, and HD95 across all three datasets, demonstrating high segmentation accuracy and robust boundary localization. It also remains competitive with fully supervised models on several metrics. Specifically, its HD95 values of 5.08 and 6.92 on AutoPET-FDG and AutoPET-PSMA, respectively, surpass those of all fully supervised models, while its remaining metrics generally reach or approach fully supervised performance.

Unlike two dimensional ICL methods such as UniverSeg and Tyche, LeSANet operates directly on three dimen-

sional PET-CT volumes, allowing it to exploit spatial lesion continuity and three dimensional morphology while reducing missed detections and discontinuous segmentation caused by insufficient interslice information. LeSANet also achieves superior overall performance compared with three dimensional ICL models such as Neuroverse and MedVerse. Its Dice score of 75.78% on AutoPET-FDG exceeds that of Iris, the second best method, by 8.63 percentage points. It achieves an HD95 of only 6.92 on AutoPET-PSMA and attains PPV and sensitivity scores of 81.86% and 89.65% on DeepPSMA, respectively. These findings indicate that support sample retrieval and lesion set level correspondence enhance the relevance of contextual information and facilitate the accurate matching and transfer of lesion priors.

The visualizations in Fig. 3 further demonstrate that LeSANet provides more complete lesion coverage while reducing false positive segmentation. Its advantage is particularly pronounced for small and extremely small lesions. When most competing models produce missed detections, sparse responses, or empty predictions, LeSANet still accurately localizes the lesions and generates segmentation results closely aligned with the ground truth annotations. This observation is consistent with its high sensitivity, indicating that its advantages arise not only from improved boundary delineation but also from enhanced detection of small and inconspicuous lesions.

### Ablation Experiment

To evaluate the effectiveness of each component, we conducted progressive ablation studies on AutoPET-FDG and AutoPET-PSMA, as reported in Tables 4 and 5. Results on DeepPSMA are provided in the appendix due to space constraints. Starting from the baseline network, we sequentially incorporated SR, SPC, CSI, and LSR. First, SR retrieves support cases with metabolic patterns similar to those of the query, thereby reducing interference from irrelevant contextual information. Consequently, Dice increases by 1.20 percentage points and HD95 decreases by 1.10 on AutoPET-PSMA. SPC then extracts semantic prototypes from support lesion regions and uses them to modulate query features. This module improves Dice and sensitivity by 0.93 and 1.32 percentage points, respectively, and reduces HD95 by 1.74 on AutoPET-FDG. It also increases PPV by 3.58 percentage points on AutoPET-PSMA, indicating that lesion prototypes enhance relevant responses while suppressing background interference.

Further incorporating CSI improves sensitivity on AutoPET-PSMA by 3.98 percentage points, demonstrating that lesion set level correspondence can anchor semantic priors to spatial locations in the query. Although sensitivity on AutoPET-FDG temporarily decreases, incorporating LSR substantially improves performance, increasing PPV and sensitivity by 3.23 and 4.60 percentage points, respectively, while reducing HD95 by 0.95. HD95 on AutoPET-PSMA also decreases by a further 1.61. These findings indicate that lesion prompting, coarse to fine gated decoding, and multi support fusion can rectify prior correspondence errors and refine segmentation boundaries.

Moreover, applying the size aware loss  $L_{\text{SA}}$  alone im-

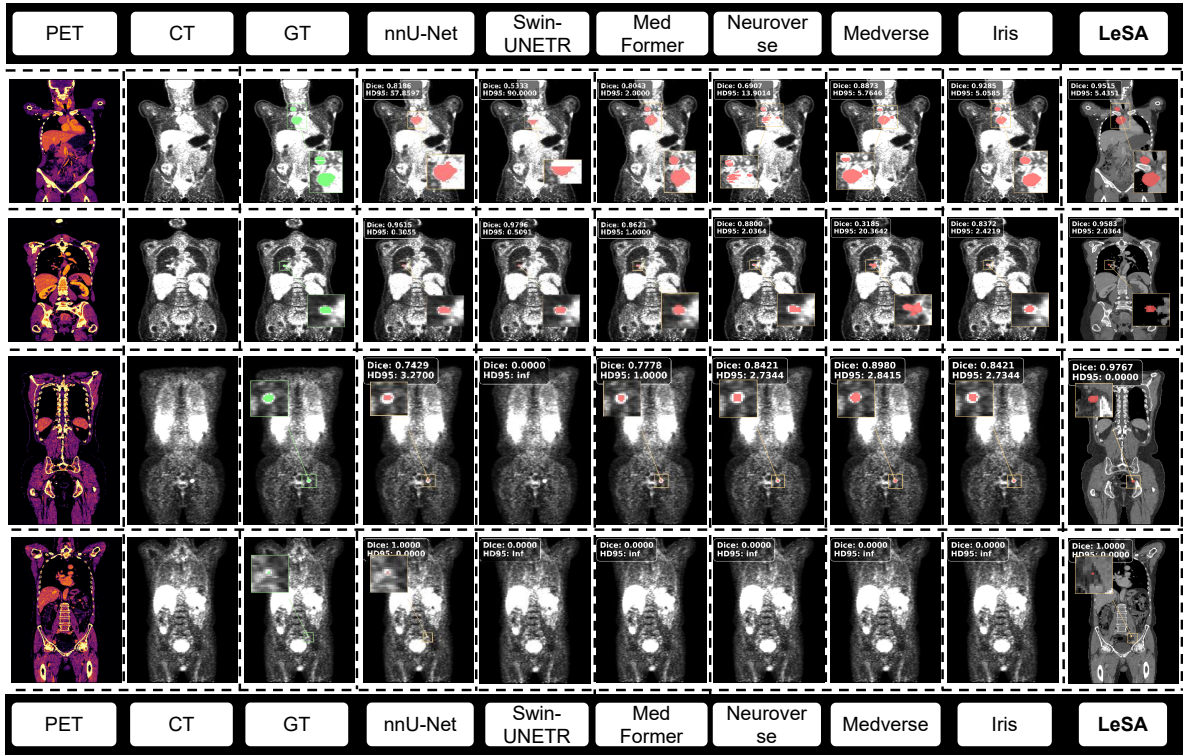


Figure 3: **Qualitative comparison of lesion segmentation results on the FDG and PSMA datasets.** The top two rows show representative FDG cases, while the bottom two rows show PSMA cases. Lesion size gradually decreases from top to bottom, covering large, medium, small, and tiny lesions. Green overlays indicate ground-truth masks and red overlays indicate model predictions. Compared with competing methods, LeSANet produces more accurate lesion delineation and demonstrates robust detection performance across lesions of diverse sizes, particularly for small and tiny lesions.

proves several metrics. Combining it with the preceding components further improves all metrics on both datasets. The complete model ultimately achieves Dice scores of 75.78% and 59.85% on AutoPET-FDG and AutoPET-PSMA, respectively, validating the complementarity of lesion prior modeling and size aware supervision.

## Conclusion

In this work, we presented LeSANet, a three-dimensional lesion-aware in-context learning framework for segmenting multiple metastatic lesions in whole-body PET-CT. Unlike conventional approaches that compress support information into a task-level representation, LeSANet models fine-grained lesion-set-level correspondences between support and query volumes. Metabolism-based support retrieval and prototype conditioning provide relevant lesion priors, cross-sample interaction anchors these priors

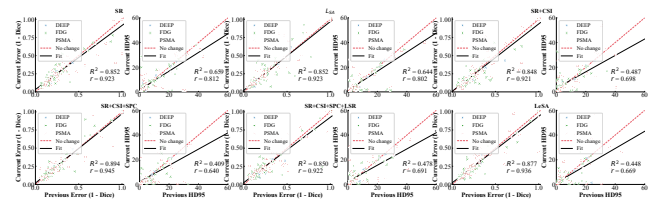


Figure 4: **Case-wise error propagation across consecutive stages of LeSANet on the DEEP, FDG, and PSMA datasets.** Each pair of panels compares the previous-stage and current-stage errors after introducing SR,  $L_{SA}$ , CSI, SPC, LSR, and the complete LeSANet framework. Left panels show Dice error ( $1 - \text{Dice}$ ), while right panels show HD95. Blue circles, green squares, and red triangles denote the DEEP, FDG, and PSMA datasets, respectively. The red dashed diagonal represents unchanged performance; points below this line indicate reduced error after the corresponding stage. The black line denotes the linear fit over all cases, with  $R^2$  and Pearson correlation coefficient  $r$  reported in each panel.



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